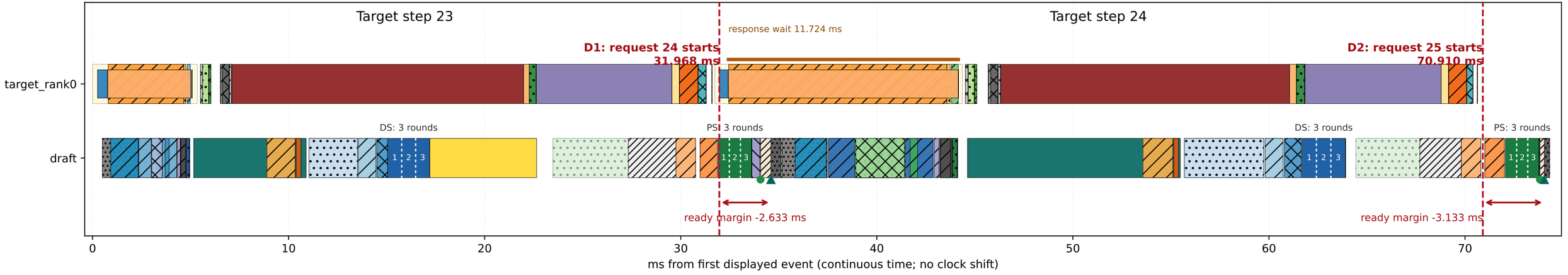
LayerSkip Llama2-13B + llama-68m | Exit 26; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | P1 hit -> P2 hit | UID 341_t0 | steps 23-24

Fresh local orion RTX4090; B1 BF16 T0.7; 12 fixed turns x 2; first10/terminal/cap excluded; GPU forward markers enabled | Exit 26; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | B=1, K1/K2 3/3

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	741	46.633	2.509	77.841
P2 hit	592	37.256	2.336	70.942
Miss	256	16.111	1.828	68.454
Overall	**1589**	**83.889**	**2.335**	**73.885**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	741	32.229	0.192	11.425	+10.848	0.168	4.263	+3.760
P2 hit	592	32.930	0.200	10.981	+10.381	0.168	4.381	+3.876
Miss	256	26.706	0.169	9.009	+8.503	0.169	2.866	+2.361
Overall	**1589**	**31.601**	**0.191**	**10.870**	**+10.296**	**0.168**	**4.082**	**+3.578**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = measured model + LM head (markers included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p1
tree_hit_parent_q_gather_p2
tree_hit_rerank_p1
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

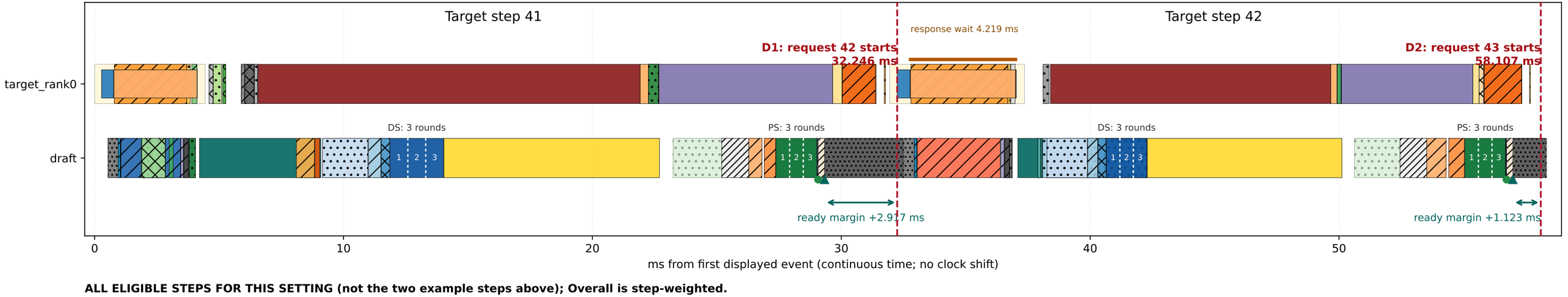
LayerSkip Llama2-13B + llama-68m | Exit 26; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | P2 hit -> Miss | UID 181_t0 | steps 41-42

Fresh local orion RTX4090; B1 BF16 T0.7; 12 fixed turns x 2; first10/terminal/cap excluded; GPU forward markers enabled | Exit 26; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | B=1, K1/K2 3/3

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	741	46.633	2.509	77.841
P2 hit	592	37.256	2.336	70.942
Miss	256	16.111	1.828	68.454
Overall	**1589**	**83.889**	**2.335**	**73.885**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	741	32.229	0.192	11.425	+10.848	0.168	4.263	+3.760
P2 hit	592	32.930	0.200	10.981	+10.381	0.168	4.381	+3.876
Miss	256	26.706	0.169	9.009	+8.503	0.169	2.866	+2.361
Overall	**1589**	**31.601**	**0.191**	**10.870**	**+10.296**	**0.168**	**4.082**	**+3.578**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = measured model + LM head (markers included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

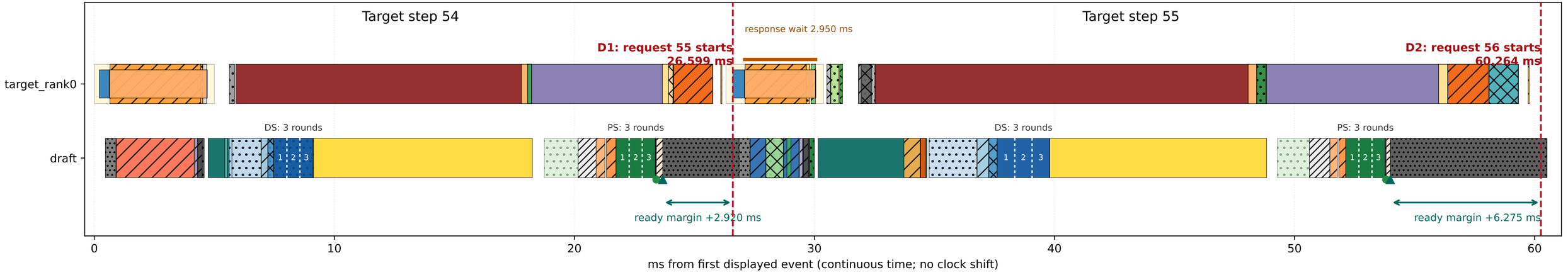
LayerSkip Llama2-13B + llama-68m | Exit 26; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | Miss -> P2 hit | UID 541_t0 | steps 54-55

Fresh local orion RTX4090; B1 BF16 T0.7; 12 fixed turns x 2; first10/terminal/cap excluded; GPU forward markers enabled | Exit 26; K1/K2 3/3; DS roots/pos 3, PS roots 9; W1/W2 9/9; C3; M40 | B=1, K1/K2 3/3

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	741	46.633	2.509	77.841
P2 hit	592	37.256	2.336	70.942
Miss	256	16.111	1.828	68.454
Overall	**1589**	**83.889**	**2.335**	**73.885**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS forward (ms)	DS compute* (ms)	DS spare* (ms)	PS forward (ms)	PS compute* (ms)	PS spare (ms)
P1 hit	741	32.229	0.192	11.425	+10.848	0.168	4.263	+3.760
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Overall	**1589**	**31.601**	**0.191**	**10.870**	**+10.296**	**0.168**	**4.082**	**+3.578**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Forward = measured model + LM head (markers included). Compute* = forward sum + spare, with current non-forward work fixed; not a safe larger-K guarantee.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_commit
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

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